



Research paper

SMEs and artificial intelligence (AI): Antecedents and consequences of AI-based B2B practices

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ABSTRACT

Development of small and medium enterprises (SMEs) is a key approach to achieving economic growth in the Middle East and successful adoption of technology is vital for SMEs' success and continuity. Artificial intelligence (AI) is part of a new generation of technologies that can facilitate competitive advantage but currently there is a lack of evidence regarding AI applications in relation to B2B SMEs in Middle East countries. Therefore, this study empirically examines antecedents to, and consequences of, successful acceptance of AI practices by B2B SMEs in Saudi Arabia. A conceptual model based on the technology-organisation-environment framework is developed which considers the impact of AI enablers and AI readiness on the acceptance of AI practices, and the impact of this on relational governance, performance, and SMEs' AI-based business customer interaction. The conceptual model was tested using structural equation modelling of survey data collected from B2B SMEs ($n = 392$). The results showed that, of the AI enablers, acceptance of AI practices was significantly influenced by both technology roadmapping and attitude but not professional expertise. Of the AI readiness variables, acceptance of AI practices was significantly influenced by infrastructure and awareness but not technicality. The acceptance of AI practices was found to significantly affect AI-enabled relational governance and performance, and SME's business customer AI-based interaction. This study provides a broader base for theoretical and practical understanding of issues related to AI practices in SMEs and the B2B sector in general.

1. Introduction

Small and medium enterprises (SMEs) account for the majority of businesses globally (World Bank, 2020) and they have “become the centrepiece of strategies to achieve inclusive growth in the Middle East, North Africa, Afghanistan and Pakistan (MENAP) region” (Lukonga, 2020, p.5). SMEs have been increasingly extending their activities within business markets in order to create more opportunities for growth and expansion (Mogaji, Soetan, & Kieu, 2020). B2B SME success largely depends on the extent to which there is an effective level of coordination and contact between SMEs and their business customers (Osterwalder & Pigneur, 2010; Rauch et al., 2018). However, the shape and nature of

B2B interactions seem to be more complex and intertwined than in the consumer market (Agnihotri, Dingus, Hu, & Krush, 2016), thus creating a need for sophisticated solutions.

Adoption of emerging technologies has contributed considerably to SMEs' ability to effectively overcome challenges, collaborate and interact with their business customers, and improve firm performance (Agnihotri et al., 2016; Brink, 2017; Dwivedi & Mustafee, 2010; Ghobakhloo & Ching, 2019; Sila & Dobni, 2012; Tan, Chong, Lin, & Eze, 2010). Artificial intelligence (AI) is a system's ability to interpret and learn from data (Aker et al., 2021; Alter, 2021; Haenlein & Kaplan, 2019), mimicking human intelligence (Rahman, Hossain, & Fattah, 2021), and it is part of a new generation of technologies that introduce

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novel approaches in the business context (Dwivedi, Hughes, Ismagilova, Aarts, et al., 2021). AI applications, such as process automation and optimisation, analytics, dynamic pricing, and prediction, can be used in diverse ways across business functions to enable organisations to obtain a variety of benefits in terms of greater revenue, efficiency, agility, productivity, and reliability, as well as better decision making and customer experience (Alsheibani, Cheung, & Messom, 2020; Dwivedi, Hughes, et al., 2021; McKinsey, 2020; Min, 2010; Rauch, Dallasega, & Unterhofer, 2019; Syam & Sharma, 2018). In a survey of executives, Ransbotham, Khodabandeh, Fehling, LaFountain, and Kiron (2019) found that 90% of respondents agree that AI presents an opportunity for their organisation and Basri (2020) has also argued that adoption of AI by SMEs can lead to increased market share and higher revenues.

Despite significant potential benefits and an annual investment volume of 55 billion dollars for the period 2015–2020 (Statista, 2021), to date there has been a slower acceptance rate of AI among both SMEs (Horváth & Szabó, 2019) and developing countries (McKinsey, 2020). Studies have emphasised the challenges faced by SMEs when implementing AI in the business process (Hansen & Bøgh, 2020; Rauch et al., 2019). According to Contissa et al. (2018), there are multiple issues associated with the acceptability and transparency of AI. There may also be, as Kozinets & Gretzel (2021, p.159) describe, “interrelated challenges of incomprehensibility, disconnection, and vulnerability” which may be difficult to navigate for B2B SMEs. A key constraint for SMEs is their lack of financial resources, which constitute the main difference between large- and small-scale organisations; large organisations will likely have greater financial resources and are thus able to invest capital to adopt technological advancements and take on greater challenges and risks (Ghobakhloo & Ching, 2019; Masood & Sonntag, 2020; Rauch et al., 2019; Vrchota, Volek, & Novotná, 2019). Moreover, it has been reported that limited access to finance is an even greater constraint for SME development in the Arab region (Stepanyan, Abajyan, Alnasaa, & Ndoye, 2019).

Given the importance placed on SMEs for economic growth in the Middle East, it is imperative to identify drivers and outcomes of adoption of new technological advancements so that these organisations succeed and remain competitive. However, there is currently a gap in the literature exploring this in the context of B2B SME acceptance of AI practices. Therefore, the questions this research aims to answer are:

RQ1: What are the main antecedents and consequences of acceptance of AI practices by B2B SMEs in Saudi Arabia?

RQ2: How could AI practices help B2B SMEs improve their business outcomes in aspects such as customer interaction, relational governance, and business performance?

To answer these questions, the next section of the paper reviews existing literature related to antecedents and consequences of AI adoption. Then, a conceptual model and hypotheses are developed identifying B2B SME AI enablers and readiness factors as antecedents of acceptance of AI practices, and consequences of acceptance in terms of SMEs' customer engagement and service experience, relational governance, and business performance. Following this, the paper explains the methods adopted to test the conceptual model empirically and, subsequently, the results of structural equation modelling using survey data collected from B2B SMEs in Saudi Arabia are reported. Finally, the paper goes on to discuss the results in the context of existing literature and the practical and theoretical implications of the findings, as well as limitations and future research directions.

2. Literature review

2.1. Technology adoption by SMEs

E-commerce has become an important medium for conducting business and numerous studies have focused on e-commerce acceptance

by B2B SMEs (e.g., Asare, Gopolang, & Mogotlhwane, 2012; Kurnia, Choudrie, Mahbur, & Alzougool, 2015; Sila & Dobni, 2012). Kurnia et al. (2015) proposed and empirically tested the impact of organisational, industry, and national readiness on the adoption of e-commerce applications, such as e-mail, electronic funds transfer, and barcodes, by B2B SMEs. They found that perceived benefits, perceived organisation resources and governance, perceived supporting services, and perceived environmental pressure had varying influences on the adoption of different e-commerce applications, demonstrating the need to examine adoption of individual applications rather than treating them as a homogenous group. Asare et al. (2012) found differences in the way B2B and B2C SMEs perceive problems in two different developing countries, but their results supported the role of power disruption and technicality in predicting the adoption of ICT applications by B2B SMEs. Based on the technology-organisation-environment (TOE) theory, Sila and Dobni (2012) explored the contextual factors influencing B2B SME e-commerce usage patterns. They found that top management support and pressure from competition were two key differences between organisations with the highest levels of e-commerce usage and integration compared with those more reluctant to adopt these technologies. Moreover, they found that firms utilising higher levels of B2B e-commerce technologies had stronger business performance, providing further support for SMEs' continued adoption of innovative technologies in order to remain competitive.

The fourth industrial revolution, or Industry 4.0, enhances the adoption of computers and automation with smart and autonomous systems, and includes technologies such as AI, Internet of Things (IoT), and blockchain (Collins, Dennehy, Conboy, & Mikalef, 2021; Hansen & Bøgh, 2020; Syam & Sharma, 2018; Verma, Sharma, Deb, & Maitra, 2021). Numerous studies have explored implementation of such technologies by SMEs although, compared to e-commerce adoption, few have focused specifically on B2B SMEs. Examining the driving forces and barriers to adopting Industry 4.0 through interviews with 26 company executives, Horváth and Szabó (2019) found that SMEs have different driving forces and barriers compared to multi-national enterprises (MNEs). For SMEs, having more limited recruitment opportunities means human resource factors are a high driving force of using Industry 4.0 as a solution. On the other hand, financial resources and profitability pose a higher barrier for SMEs since they often have fewer financial resources to invest in new technologies compared to MNEs. In a survey of UK manufacturing SMEs ($n = 271$), Masood and Sonntag (2020) found that perceived benefits of implementing Industry 4.0 technologies was affected by company size and attitude towards Industry 4.0, whereas perceived challenges of implementing Industry 4.0 was affected by manufacturing complexity. Open text responses regarding how SMEs could prepare for future manufacturing systems and technologies suggested training, financial support, and investment in people were key. In a survey of manufacturing SMEs in Iran ($n = 183$) and Malaysia ($n = 177$), Ghobakhloo and Ching (2019) found that strategic roadmapping for manufacturing digitalization is the most significant determinant of Smart Manufacturing-related Information and Digital Technologies (SMIDT) adoption. They found that AI SMIDT adoption specifically is affected by perceived costs, perceived value, perceived compatibility, and IDT knowledge competency.

2.2. AI adoption by B2B SMEs

Research has paid increasing attention over recent years to issues of AI adoption by organisations (Table 1). The majority of these studies have focused on antecedents rather than consequences of adoption. There have been both qualitative (e.g., Bedué & Fritzsche, 2021; Jöhnk, Weißert, & Wyrski, 2021) and quantitative studies that have examined organisations' AI adoption intentions (e.g., Alsheibani et al., 2020; Cao, Duan, Edwards, & Dwivedi, 2021; Chatterjee, Rana, Dwivedi, & Baabdullah, 2021; Ghobakhloo & Ching, 2019). Management or leadership support has been found to be an important facilitator of AI adoption in

Table 1

Studies that have examined adoption and implementation of AI by organisations.

Study	Focal technology	Methodology	Key findings
Alsheibani et al. (2020)	AI	Questionnaire survey of organisations in Australia ($n = 208$), primarily large organisations.	Top management support is the strongest determinant of AI adoption. Compatibility, relative advantage, organisational readiness, managerial obstacles, and government regulations were also found to significantly affect AI adoption.
Bag et al. (2021)	Big data analytics powered AI (BDA-AI)	Questionnaire survey of automotive organisations in South Africa ($n = 219$).	Tangible resources and workforce skills have a positive effect on adoption of BDA-AI. Adoption of BDA-AI has a positive effect on sustainable manufacturing and circular economy capabilities.
Basri (2020)	AI-assisted social media marketing	Questionnaire survey of SMEs in Saudi Arabia ($n = 78$).	AI-assisted social media marketing has a positive effect on both effective business management and SME performance.
Bedué and Fritzsche (2021)	AI	Qualitative semi-structured interviews with 12 decision-makers from different industries and organisation sizes.	Access to knowledge, transparency, explainability and certification, as well as self-imposed standards and guidelines, to be important for reducing uncertainty, increasing trust in AI, and enhancing adoption intentions.
Cao et al. (2021)	AI for organisational decision-making	Questionnaire survey of managers in medium and large organisations in the UK ($n = 269$).	Managers' attitudes towards AI are positively influenced by performance expectancy and effort expectancy but negatively influenced by personal wellbeing concern and perceived threat. Intention to use AI is positively influenced by attitude but negatively influenced by personal development concern and personal wellbeing concern.

Table 1 (continued)

Study	Focal technology	Methodology	Key findings
Chatterjee, Rana, Dwivedi, and Baabdullah (2021), Chatterjee, Rana, Tamilmani, & Sharma, 2021	AI	Questionnaire survey of top- and middle-level professionals of companies where AI is being adopted or contemplated in India ($n = 340$).	Perceived usefulness (PU) of AI is positively affected by organisational competency, complexity, readiness, and compatibility, as well as partner support, competitive advantage, and perceived ease of use (PEOU). PEOU of AI is positively affected by organisational complexity and competitive advantage. Both PEOU and PU positively affect intention to adopt AI but both of these effects are moderated by leadership support.
Dubey, Gunasekaran, Childe, Bryde, et al. (2020)	Big data analytics powered by artificial intelligence (BDA-AI)	Questionnaire survey of manufacturing firms from eight different industries in India ($n = 256$).	Entrepreneurial orientation positively affects BDA-AI and operational performance, although environmental dynamism moderates both relationships. BDA-AI also positively affects operational performance.
Ghobakhloo and Ching (2019)	Smart Manufacturing-related Information and Digital Technologies (SMIDT)	Questionnaire survey of manufacturing SMEs in Iran ($n = 183$) and Malaysia ($n = 177$).	Found that strategic roadmapping for manufacturing digitalization is the most significant determinant of SMIDT adoption. AI SMIDT adoption specifically is affected by perceived costs of SMIDT, perceived value of SMIDT, perceived compatibility of SMIDT, and IDT knowledge competency.
Hansen and Bøgh (2020)	AI and IoT	Systematic literature review of IoT and AI for industrial SMEs.	Found that more research is needed on the subject of IoT and AI in SMEs to unlock the potential of Industry 4.0.
Jöhnk et al. (2021)	AI	Qualitative in-depth interviews with 25 AI experts.	Derived 18 AI readiness factors in five broad categories of strategic alignment, resources,

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Table 1 (continued)

Study	Focal technology	Methodology	Key findings
Rahman et al. (2021)	AI	Questionnaire survey of managers from large and medium-sized manufacturing and service-intensive firms in Bangladesh ($n = 250$).	knowledge, culture and data. AI adoption enhances the relationships between firms' marketing analytics capability and holistic marketing decision-making, and firms' marketing analytics capability and firms' competitive marketing performance.

organisations (e.g., [Alsheibani et al., 2020](#); [Chatterjee, Rana, Dwivedi, & Baabdullah, 2021](#); [Jöhnk et al., 2021](#)), as has compatibility (e.g., [Alsheibani et al., 2020](#); [Ghobakhloo & Ching, 2019](#); [Jöhnk et al., 2021](#)). Of the few studies that have focused on consequences of AI adoption, [Basri \(2020\)](#) found that AI-assisted social media marketing has a positive effect on both effective business management and SME performance, and ([Rahman et al. \(2021\)](#)) found that AI adoption enhances the relationships between firms' marketing analytics capability and holistic marketing decision-making, and firms' marketing analytics capability and firms' competitive marketing performance.

Within the B2B context, [Chatterjee, Rana, Tamilmani, and Sharma \(2021\)](#) have also argued that AI-based CRM applications could help business organisations to enhance their business performance based on two models (institutional theory and the resource-based view (RBV)). Their empirical results largely supported the importance of organisation fit, individual fit, and implementation process quality as key antecedents of successful implementation of AI-based CRM for B2B. Further, [Chatterjee, Rana, Tamilmani, and Sharma \(2021\)](#) confirmed that AI-based CRM can predict the business performance via the mediating impact of B2B engagement, employee experience, and B2B information process.

[Hansen and Bøgh \(2020\)](#) argue that more research is needed on the subject of IoT and AI in SMEs to unlock the potential of Industry 4.0. This is especially in the light of the fact that practitioners all over the world are looking forward to finding the best AI practices which are able to provide a better service and unique experience for customers, thus improving institutional performance ([Kushwaha, Kar, & Dwivedi, 2021](#); [Verma et al., 2021](#)). However, in their systematic review, [Verma et al. \(2021\)](#) pointed out that there is a lack of studies that have tested the related issues of AI acceptance in the marketing field. Therefore, practitioners suffer from a lack of full knowledge of the important aspects for the success of using this technology, which in turn, consumes more time, effort and investment ([Kushwaha et al., 2021](#)).

While progress has been made in examining the antecedents and consequences of AI acceptance, there are still some limitations. For example, research participants tend to be from primarily large organisations (e.g., [Alsheibani et al., 2020](#); [Cao et al., 2021](#); [Rahman et al., 2021](#)), few studies have explored both antecedents and consequences of AI adoption simultaneously (e.g., [Bag, Pretorius, Gupta, & Dwivedi, 2021](#)), and there is little explicit consideration of B2B SMEs specifically. This, in turn, presents research gaps worthy to be examined based on a solid theoretical foundation covering the most important antecedents and consequences of AI-practices and suitable to the nature of B2B SMEs. Therefore, this study seeks to contribute to these gaps by considering both antecedents and consequences of AI practices for B2B SMEs in the context of Saudi Arabia.

3. Theoretical background, proposed conceptual model and hypotheses

A conceptual model is primarily used to create a relationship among all the variables associated with a research approach. [Tornatzky, Fleischer, and Chakrabarti \(1990\)](#) evaluated the technology-organisation-environment (TOE) model for suggesting significant steps when adopting any technological aspect in the business process. As it provides a comprehensive picture regarding the main factors that could determine the adoption of new systems from the organisational perspective, TOE has been extensively used as a theoretical base by several studies over different contexts such as smart cities ([Ullah, Qayyum, Thaheem, Al-Turjman, & Sepasgozar, 2021](#)), green innovation ([Zhang, Sun, Yang, & Wang, 2020](#)), RFID ([Aboelmaged & Hashem, 2018](#)), social commerce ([Abed, 2020](#)), big data analytics ([El-Haddadeh, Osmani, Hindi, & Fadlalla, 2020](#)), IoT ([Schmitt, Mladenow, Strauss, & Schaffhauser-Linzatti, 2019](#)), and big data in the B2B context ([Sun, Hall, & Cegielski, 2020](#)).

According to the TOE model, factors predicting the successful implementation and adoption of emerging systems fall into three key dimensions: technology, organisation, and environment ([Tornatzky et al., 1990](#)). Technological dimensions pertain to these related technologies (software and hardware) that are available inside or outside the organisation and can facilitate the adoption of AI practices. In the current study, a number of aspects fall under the technological component of the TOE model: AI-based technology roadmapping, infrastructure, and technicality. The second component in the TOE model is organisation which incorporates elements such as the size of the organisation, organisational chart, human resources, organisational culture, and departmental interconnections ([Cruz-Jesus, Pinheiro, & Oliveira, 2019](#); [Junior, Oliveira, & Yanaze, 2019](#)). These aspects have been largely approved as key determinants of successful adoption of new systems (e.g., [Junior et al., 2019](#); [Venkatesh & Bala, 2012](#)). AI-based professional expertise, attitude, and awareness were proposed in the current study to capture the organisational component of TOE. The third TOE component relates to the environment in which all business activities are conducted and includes industry, suppliers, competitors, and stakeholders ([Martins, Oliveira, Thomas, & Tomás, 2019](#); [Tornatzky et al., 1990](#)). As the main context of this study is B2B, business customers were included under the environment component which is reflected by two main aspects in the current study: service experience with AI and AI-based customer engagement.

In this paper, multiple variables measuring antecedents and consequences of AI are adopted so that thorough conclusions can be drawn. [Dwivedi, Hughes, et al. \(2021\)](#) researched the opportunities provided by AI for business organisations and argued that the nature or organisational culture of SMEs facilitates adoption of technological elements in business premises. [Ayyagari, Demircug-Kunt, and Maksimovic \(2014\)](#) claimed that a significant orientation among SMEs would be required to enhance their existing AI framework. They strongly recommended that SMEs should run a continuous system for AI because a steady system can be updated to a better version of a previous one. According to [Baker \(2012\)](#), a company's adoption of any sort of technological tool is highly dependent on its readiness, organisational behaviour, and organisational environment. These three aspects are essential for understanding whether organisations are ready to adopt technology ([Baker, 2012](#); [Tornatzky et al., 1990](#)). In the present research context, this conceptual model would enable a strong interrelation to be made among all the B2B and AI variables of Saudi SMEs ([Baker, 2012](#); [Tornatzky et al., 1990](#)). The conceptual model for analysing this research topic is presented in [Fig. 1](#). Definitions of the main constructs are also provided in [Table 2](#).

3.1. SMEs' AI enablers and AI practices

SMEs adopt modern technologies like AI in order to enhance business performance ([Dwivedi, Hughes, et al., 2021](#)). According to [Sohn and](#)

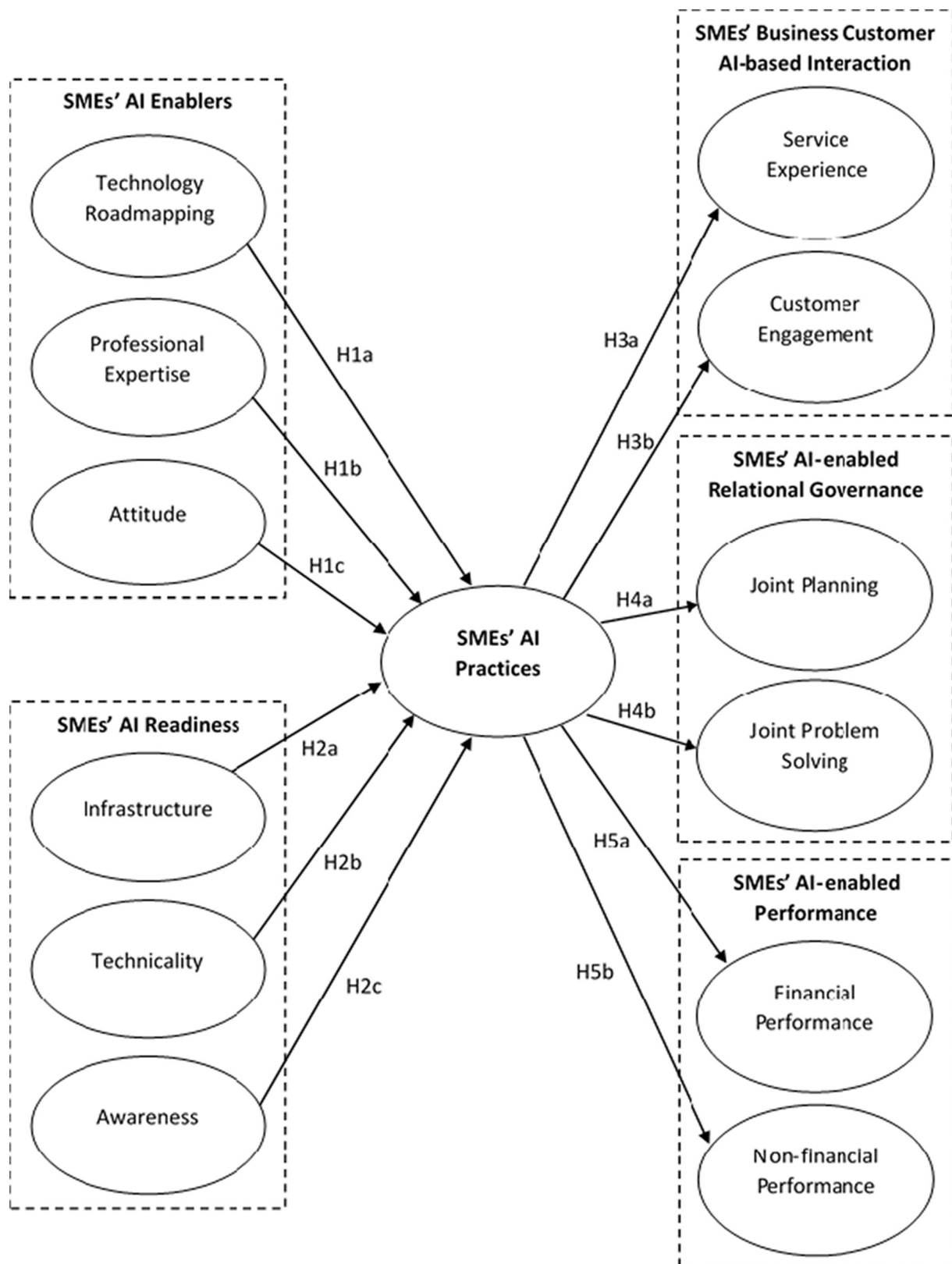


Fig. 1. Proposed conceptual model (adapted from Tornatzky et al., 1990).

Kwon (2020), various driving forces allow business organisations to adopt such technologies, and these driving forces are AI enablers. Three types of AI enabler can influence business operations: technology roadmapping, professional expertise, and attitude. Technology

roadmapping is a strategic planning technique for research, development and implementation of new technologies. In an analysis of the adoption of Smart Manufacturing-related Information and Digital Technologies (SMIDT) in the business operations of Malaysian and

Table 2
Definitions of constructs.

Construct	Definition	Source
Technology roadmapping	A strategic planning process to objectively identify the technological alternatives available to satisfy organisational objectives.	Ghobakhloo and Ching (2019)
Professional expertise	The extent of SME employees' experience, knowledge, and skills with AI-based applications.	Lin and Lin (2008)
Attitude	Overall evaluation of AI-based applications.	Cao et al. (2021)
Infrastructure	The systems, software, and IT complementary resources that are necessary for AI-based applications.	Lin and Lin (2008)
Technicality	The extent to which SME employees perceive AI-based applications as technically user friendly.	Sigala (2006)
Awareness	The degree of SME employees' knowledge about AI-based applications.	Alsheikh and Bojei (2014)
SMEs' AI practices	The extent of acceptance of AI applications by SMEs.	Ghobakhloo and Ching (2019)
Service experience	Functional clues to the technical quality of the AI service offering, indicating the reliability and competence of the service.	Prentice and Nguyen (2020)
Customer engagement	A cognitive and affective commitment to an active relationship with the organisation as a result of AI.	Mollen and Wilson (2010)
Joint planning	The extent to which business partners can jointly plan and design future contingencies and consequential duties and responsibilities in a relationship as a result of AI.	Claro, Hagelaar, and Omta (2003)
Joint problem solving	The extent to which business partners can mutually solve their disagreements and problems as a result of AI.	Claro et al. (2003)
Financial performance	The financial results of AI integration that can be quantitatively measured such as profitability, sales volume, and return on investment.	Diffley and McCole (2015); Hernaus, Pejić Bach, and Bosilj Vukšić (2012); Jain, Khalil, Johnston, and Cheng (2014)
Non-financial performance	The non-financial results of AI integration in terms of customer satisfaction, loyalty, and competitive advantage.	Diffley and McCole (2015); Hernaus et al. (2012); Jain et al. (2014)

Iranian SMEs, Ghobakhloo and Ching (2019) found that strategic roadmapping for manufacturing digitalization is the most significant determinant of SMIDT adoption. In line with this, Bangalore Seetharam (2020) argued that a digital AI-based roadmap would be required for effective implementation of AI in the business process. The author researched this issue by concentrating on an IT firm that wanted to adopt AI, finding that the firm faced serious challenges due to an inappropriate technological framework. For this reason, Bangalore Seetharam (2020) recommended a 'value generation team' for the organisation that could create the roadmap for AI implementation. Accordingly, this study proposes the following hypothesis:

H1a. Technology roadmapping has a positive influence on B2B SMEs' AI practices.

In a qualitative study with 25 AI experts, Jöhnk et al. (2021) noted the importance of knowledge and culture as prerequisite factors to successful AI implementation. Similarly, in a quantitative study, Bag et al. (2021) found that workforce skills positively affect adoption of big

data analytics powered by AI. Within the B2B SME sector, it has been supported that having staff with adequate levels of professional expertise is important for ensuring the success of technological transformation and technology adoption (Alsudairi & Dwivedi, 2010; Kurnia et al., 2015; St-Pierre & Audet, 2011). Cantarello, Martini, and Nosella (2012) demonstrated that the performance of the development process largely depends on the extent to which employees enjoy professional expertise in terms of the skills and cognitive knowledge required. Furthermore, professional expertise has been widely shown to play a significant role in shaping the innovation diffusion process across the business sector (Lin & Lee, 2005; Lin & Lin, 2008; Zhu, Kraemer, & Xu, 2003). Accordingly, B2B SMEs that have staff with high technical skills and capabilities are more likely to successfully implement AI practices, which then positively impacts their business customer interactions and business performance (Agostini & Nosella, 2016; Ghobakhloo & Ching, 2019). Accordingly, this study suggests the following hypothesis:

H1b. Professional expertise has a positive influence on AI practices in B2B SMEs.

In the context of AI, attitude explains the positive or negative feelings individuals have towards the technology (Cao et al., 2021). To implement a technology like AI, organisational leaders have to be sufficiently positive about it (Anandarajan, 2002). Lee and Runge (2001) studied the perception and behaviour of leaders towards the adoption of new IT in the business process. They found that leaders of small firms must have a positive attitude towards the technology for its successful implementation. A study by Masood and Sonntag (2020) found that attitude towards Industry 4.0 had the strongest effect on perceived benefits. More recently, Cao et al. (2021) found that managers' attitudes are the strongest predictor of intention to use AI for organisational decision-making. Hence, the following hypothesis is proposed:

H1c. Attitude towards AI has a positive influence on B2B SMEs' AI practices.

3.2. SMEs' AI readiness and AI practices

SMEs' AI readiness refers to organisations' capabilities to support AI practices (Alsheibani et al., 2020). In the current study three aspects are associated with AI readiness: AI infrastructure, awareness, and technicality. It has been widely argued that technical infrastructure is a fundamental prerequisite for a successful digital transformation process (e.g., Kowtha & Choon, 2001; Lin & Lee, 2005; Lin & Lin, 2008). Recent research has found that IT infrastructure is one of the key AI readiness factors, specifically three underlying elements of IT infrastructure: data storage, networking, and scalable computing power capabilities (Jöhnk et al., 2021). Moreover, Alsheibani et al. (2020) found that technology infrastructure and data structure were critical components of organisational readiness which positively influenced AI adoption. Based on these findings, this study hypothesises that:

H2a. AI infrastructure has a positive influence on B2B SMEs' AI practices.

In addition to AI infrastructure, AI technicality is also essential. Technicality can be operationalised as the extent to which the targeted system (AI) is perceived to be uncomplicated for users (Sigala, 2006). If users perceive that a system is free of physical and mental effort then they are more likely to be inclined to use it; likewise, a system that is perceived to be technically difficult will be less likely to be used (Sohn & Kwon, 2020). In the context of AI adoption and implementation by organisations, several studies have found factors relating to user friendliness (e.g. ease of use, effort expectancy) to have a significant positive effect on adoption (e.g. Cao et al., 2021; Chatterjee, Rana, Dwivedi, & Baabdullah, 2021; Chatterjee, Rana, Tamilmani, & Sharma, 2021). Therefore, this study hypothesises that:

H2b. AI technicality has a positive influence on AI practices in B2B

SMEs.

Awareness “implies knowledge about a technology, its benefits, and risks that are key factors in the voluntary use of systems” (Alsheikh & Bojei, 2014, p.212). In the context of SMEs in the EU, Delponte and Tamburrini (2018) claimed that AI awareness among organisational stakeholders is a significant determinant of AI readiness and that organisational stakeholders must be given appropriate knowledge regarding AI so that they can use the technology to benefit the organisation. EU Member States have launched various AI awareness programmes and digital classrooms for consumers and SME employees (Delponte & Tamburrini, 2018). Thus, it could be argued that the acceptance of AI practices by B2B SMEs largely depends on the extent to which they know about AI applications and have enough information about the benefits of such new systems. Hence, this study hypothesises that:

H2c. AI awareness has a positive influence on B2B SMEs' AI practices.

3.3. SMEs' AI practices and business customer AI-based interaction

AI is an innovation that can contribute to all aspects of business customer interactions (Dwivedi, Hughes, et al., 2021). Practically, AI has changed the business dimension and allowed organisations to be competitive by establishing a special service experience, customer engagement, and, accordingly, brand value in the marketplace (Ameen, Tarhini, Reppel, & Anand, 2021; Chen, Guo, Gao, & Liang, 2020; Haenlein & Kaplan, 2019; Prentice & Nguyen, 2020; Prentice, Weaven, & Wong, 2020; Wang, Teo, & Janssen, 2021). According to Prentice and Nguyen (2020), implementation of AI has changed how organisations serve and interact with their customers. In this respect, the critical role of AI practices in shaping B2C customer experience has recently received considerable attention by marketing researchers (e.g., Ameen et al., 2021; Prentice & Nguyen, 2020). With the help of AI-based practices, organisations are more able to efficiently serve their customers anytime and anywhere. This, in turn, not only changes the customer experience but also accelerates the organisation's performance (Prentice & Nguyen, 2020). Morosan and Bowen (2018) demonstrated how AI applications can process a huge amount of customer data in the service sector, and, accordingly, how they can enrich the service experience by providing a high level of customisation. AI provides companies with a great opportunity to know their customers' needs, wants, expectations, and buying behaviour more accurately. Thus, companies can enhance and create added value in the customer experience more than ever before (Ameen et al., 2021). Accordingly, it is hypothesised that:

H3a. B2B SMEs' AI practices have a positive influence on service experience with AI.

As well as benefitting the service experience, AI applications can also contribute to how much the customer engages with suppliers (Prentice et al., 2020). Customers are more likely to be engaged with their suppliers when there is an interactive experience (Vivek, Beatty, & Morgan, 2012). Given the innovative features of AI that could provide customers with more interactivity with suppliers, customer engagement is more attainable in comparison with traditional mechanisms used in this respect (Prentice & Nguyen, 2020). The crucial role of AI in customers' engagement has been recently shown by Prentice et al. (2020) who revealed that both AI information satisfaction and AI system satisfaction significantly contribute to the level of customer engagement. Prentice and Nguyen (2020) also provided further evidence supporting the crucial role of AI-based service experience in customers' tendency to engage with service providers. Ayyagari et al. (2014) argued that if SMEs can implement AI technology well then interaction between organisational stakeholders would be smooth. They also argued that AI implementation would be the best approach to engage customers directly with organisational activities. Based on the above discussion, it is proposed that:

H3b. B2B SMEs' AI practices have a positive influence on AI-based customer engagement.

3.4. SMEs' AI practices and relational governance

It has been commonly argued that relational governance largely relies on the extent to which business partners collaborate, share knowledge, co-create value, and depend on each other (Janssen, van der Voort, & Wahyudi, 2017). This would not be possible without sophisticated systems like AI (McAfee, Brynjolfsson, Davenport, Patil, & Barton, 2012). Practically, AI helps business partners to jointly collect, organise, and analyse high quality big data. This, in turn, improves governance and the decision-making process (Janssen et al., 2017; Nasseef, Baabdullah, Alalwan, Lal, & Dwivedi, 2021). Haenlein and Kaplan (2019) argued that AI has the potential to collect various data and analyse them to the benefit of the organisation. Business organisations traditionally had to carry out a manual survey to understand consumer demands and opinions in the target market but now they can easily analyse data and understand customer perceptions with the help of AI; accordingly, they can mutually solve problems and prepare future plans with their business customers (Mogaji et al., 2020; Prentice & Nguyen, 2020). Dwivedi, Hughes, et al. (2021) argued that AI has given enough flexibility to business partners to validate large data sets on any business strategy. This then facilitates business partners in delivering the highest quality experience to their target business customers in terms of joint planning and problem solving (Haenlein & Kaplan, 2019). Dubey et al. (2020) recently reported that AI practices will sustain the business partners' alignment by reconfiguring their resources to meet unexpected situations. Accordingly, this study proposes that:

H4a. B2B SMEs' AI practices have a positive influence on joint planning.

H4b. B2B SMEs' AI practices have a positive influence on joint problem solving.

3.5. SMEs' AI practices and AI-enabled performance

Innovation and smart systems have been widely demonstrated in the business literature to be key mechanisms contributing to both financial and non-financial performance, and this is no different for B2B SMEs (Li & Mitchell, 2009; Merrilees, Rundle-Thiele, & Lye, 2011; Weerakkody, Dwivedi, & Kurunananda, 2009; Weerawardena, O'Cass, & Julian, 2006). According to Allam and Dhunny (2019), technology is upgraded based on the needs and demands of its users. AI is one of the finest inventions of modern data science (Contissa et al., 2018). Dwivedi, Hughes, et al. (2021) and Grover, Kar, and Dwivedi (2020) claimed that AI has the potential to increase business performance and operations. Therefore, this study has adopted the proposition that acceptance of AI practices within innovative systems will positively impact the business performance of B2B SMEs.

Highly complex and diversified information is much easier to process and interpret by facilitating AI practices (Chen et al., 2020; Dubey et al., 2020; Dubey, Gunasekaran, Childe, Fosso Wamba, et al., 2019). Accordingly, business organisations are better able to be sure about their decisions relating to market demand, resource flow, and capacities. This, in turn, will mitigate the level of inventory and stock that organisations need to meet unexpected customer demand, thereby positively impacting financial performance. This assumption has recently been supported by Dubey et al. (2020) who validated a strong and significant relationship between AI practices and operational performance. In a qualitative study designed to evaluate the impact of AI on the business performance of retail organisations, Bangalore Seetharam (2020) found that SMEs have been able to improve their supply chain management during the lockdown period with the help of AI technology. The author claimed that a secured IT network between vendor and client organisations makes the supply chain process smooth and efficient, and that SMEs can

get huge benefits if they adopt AI in dealing with other firms. A recent study conducted in Saudi Arabia by [Basri \(2020\)](#) confirmed that AI-assisted social media marketing has a positive effect on both effective business management and SME performance. Furthermore, based on a questionnaire survey of managers from large and medium-sized manufacturing and service-intensive firms in Bangladesh ($n = 250$), [Rahman et al. \(2021\)](#) found that AI adoption enhances the relationships between firms' marketing analytics capability and firms' competitive marketing performance. Accordingly, it is hypothesised that:

H5a. B2B SMEs' AI practices have a positive influence on financial performance.

H5b. B2B SMEs' AI practices have a positive influence on non-financial performance.

4. Methodology

To empirically test the proposed hypotheses, and since validated measurement scales were already available to be adapted from previous studies, a survey method was employed using a structured questionnaire. Measurement items (Appendix) were selected based on their alignment with the definitions of constructs outlined in [Table 2](#). Items were measured using a five-point Likert scale anchored 1 *strongly disagree* to 5 *strongly agree*. The questionnaire was translated into Arabic based on the back-translation approach proposed by [Brislin \(1976\)](#). The Arabic version of the questionnaire was also inspected by several experts in the area of information systems, digital marketing, industrial marketing, and SMEs. These experts approved the validity of the scale items and the accuracy of the translation process.

The target population was B2B SME CEOs, CIOs, general managers, senior managers, marketing managers, and owners in Saudi Arabia. The targeted participants worked in wholesaling, manufacturing, agriculture, information technology and social media, and construction industries that provide services and products to business customers. These sectors were selected based on what has been suggested by [Merrilees et al. \(2011\)](#). Firstly, a pilot study with 25 such participants was undertaken to ensure scale reliability. The Cronbach's coefficient alpha values for all constructs were above 0.70, as recommended by [Nunnally \(1978\)](#). The questionnaire was then distributed to 500 target participants from July 2020 to the end of September 2020. Different methods were used to reach the targeted respondents. For example, contact was initially made with public relations offices in these companies to reach the target respondents. Also, respondents were contacted through their publicly available email address or through other means, such as professional social network sites. Moreover, field visits were made to suitable SMEs to request their participation.

SPSS v23 and AMOS were used for data analysis. Structural equation modelling (SEM) was used to analyse the proposed conceptual model, combining multiple regression with factor analysis, while also allowing for both observed and latent variables to be analysed simultaneously, and providing overall model fit statistics ([Tabachnick & Fidell, 2007](#)). In accordance with the recommendation of a two-stage analytical procedure ([Anderson & Gerbing, 1988](#)) confirmatory factor analysis was conducted in AMOS using Maximum Likelihood Estimation, which was then followed by path analysis of the structural relationships.

5. Results

5.1. Respondents' demographic characteristics

[Table 3](#) provides full descriptive details of the respondents' characteristics. The majority of the respondents were male (75.3%). More than 40% of the sample were between the age of 31–40, with the next most represented age category being 25–30 (19.13%). There was quite an even distribution of respondents across the different role profiles, although 17.6% were CIOs and 17% were marketing managers. Nearly

Table 3

Respondents' descriptive characteristics.

Characteristic	Category	Number of respondents (n = 392)	Percentage (%)
Gender	Male	295	75.3
	Female	97	24.7
Age	18–24	44	11.22
	25–30	75	19.13
	31–40	177	45.1
	41–50	51	13.01
	51–60	36	9.18
	60+	9	2.29
Role within the organisation	CEO	56	8.00
	CIO	69	17.6
	General manager	51	13.0
	Senior manager	45	11.4
	Marketing manager	67	17.0
	Owner	63	16.0
Education level	Operational manager	41	10.4
	High school	31	7.9
	Diploma	42	10.7
	Bachelor	234	59.6
	Master	67	17.09
	PhD	13	3.3
Industry	Other	5	1.2
	Wholesaling	127	32.3
	merchandising		
	Manufacturing	43	10.9
	Construction	22	5.6
	Service	141	35.9
	Agriculture	19	4.8
	Information technology and social Media	40	10.2
	Marketing		
	1–25	326	83.1
Number of employees	26–59	50	12.7
	60–100	16	4.01

four-fifths of participants stated that they hold a bachelor's degree or higher. In terms of industry, the majority of the participating SMEs were either from the services sector (35.9%) or wholesaling merchandising (32.3%). Finally, more than 80% of the SMEs in the sample had fewer than 25 employees.

5.2. Descriptive statistics of the measurement items

The mean and standard deviation (SD) values were calculated for the 51 scale items used in the questionnaire ([Table 4](#)). The mean values of almost all scale items were greater than 3, indicating that respondents tended to agree with the majority of the statements. However, the majority of the scale items also had SD values above 1 indicating some variation in respondents' opinions. This variation could be attributed to the fact that AI applications are still new in the B2B SME context in Saudi Arabia. Considering demographic characteristics ([Table 3](#)), the respondents were from different sectors (i.e. wholesaling merchandising, manufacturing, construction, service, agriculture) and had different roles within the organisations (i.e. CEO, CIO, General manager, Senior manager, Marketing manager, Owner, Operational manager). Therefore, people are more likely to have different opinions, perception, and awareness towards such emerging applications.

Participants tended to have positive attitudes towards (mean = 3.45, SD = 1.03), and a good level of awareness of (mean = 3.72, SD = 1.17), AI applications. There was some level of variation in participants' opinions of the strength of AI technology roadmapping of their organisation (mean = 3.06; SD = 1.21). However, technicality items were more positively rated by respondents (mean = 3.24; SD = 1.02), suggesting it would not be difficult to use AI applications in their organisation. Participants tended to agree that their organisations had good infrastructure for AI (mean = 3.74, SD = 0.93), although there was slightly more variation in opinion on AI related professional expertise

Table 4
Descriptive statistics of scale items.

Construct	Item	Item Mean	Item SD	Mean	SD
Technology roadmapping	TRM1	3.35	1.26	3.06	1.212
	TRM2	3.56	1.18		
	TRM3	3.44	1.18		
	TRM4	2.61	1.26		
	TRM5	2.31	1.25		
Awareness	AWR1	3.69	1.15	3.72	1.17
	AWR2	3.71	1.15		
	AWR3	3.69	1.16		
	AWR4	3.78	1.24		
Attitude	ATT1	3.66	1.00	3.45	1.03
	ATT2	3.32	1.04		
	ATT3	3.15	1.10		
	ATT4	3.51	1.03		
	ATT5	3.62	0.95		
Technicality	T.C1	3.23	1.06	3.24	1.02
	T.C2	3.29	0.99		
	T.C3	3.02	1.06		
	T.C4	3.44	1.00		
Professional expertise	PRE1	3.34	1.18	3.23	1.19
	PRE2	3.36	1.14		
	PRE3	3.00	1.26		
Infrastructure	INFR1	3.70	0.94	3.74	0.93
	INFR2	3.80	0.92		
	INFR3	3.73	0.92		
SMEs' AI practices	P.AI1	3.77	1.09	3.72	1.13
	P.AI2	3.73	1.11		
	P.AI3	3.67	1.17		
	P.AI4	3.70	1.14		
Customer engagement	ENG1	3.81	1.07	3.93	0.99
	ENG2	3.83	1.04		
	ENG3	3.84	1.06		
	ENG4	4.26	0.80		
Service experience	EX1	4.01	1.04	3.72	1.00
	EX2	3.81	1.03		
	EX3	3.88	1.05		
	EX4	3.86	0.98		
	EX5	3.85	0.98		
	EX6	3.75	1.02		
	EX7	3.72	1.01		
Non-financial performance	N-F1	3.73	1.07	3.56	1.13
	N-F2	3.40	1.18		
	N-F3	3.55	1.13		
Financial performance	FIN1	3.62	1.09	3.56	1.11
	FIN2	3.56	1.11		
	FIN3	3.51	1.13		
Joint problem solving	R.GPS1	3.73	1.12	3.70	1.15
	R.GPS2	3.69	1.14		
	R.GPS3	3.68	1.17		
Joint planning	R.GP1	3.51	1.10	3.48	1.09
	R.GP2	3.51	1.08		
	R.GP3	3.42	1.08		

(mean = 3.23; SD = 1.19). As for the consequences of AI practices, participants highly valued the effective use of AI applications for business activities in terms of customer interaction, business performance, and relational governance. For instance, respondents quite strongly indicated that AI practices help them to be more engaged with their business customers (mean = 3.93, SD = 0.99). Similarly, respondents felt that AI helped with service experience (mean = 3.72; SD = 1.00). Financial performance (mean = 3.56; SD = 1.11) and non-financial performance (mean = 3.56; SD = 1.13) related to AI applications were rated positively by participants as well. Finally, participants tended to perceive that AI applications help shared planning (mean = 3.48; SD = 1.09) and problem solving (mean = 3.70; SD = 1.15).

5.3. Common method bias

To rule out issues of common method bias, this study used Harman's single factor test, which eradicates concern about common method bias if the variance accounted by the first factor is not higher than 50% (Podsakoff, MacKenzie, & Lee, 2003). Fifty-one scale items for thirteen

latent constructs (i.e., FIN, TRM, AWR, ATT, INFR, PRE, T.C, EX, R.GP, P.AI, R.GPS, ENG, and N-F) were loaded for exploratory factor analysis. Cumulative variance extracted was 42.829% for the first factor, hence, the current study data does not have concerns regarding common method bias.

5.4. Structural equation modelling (SEM) analysis

Fourteen latent constructs and 51 scale items were included in the SEM analysis. In the first stage, the measurement model was tested for goodness of fit and construct reliability and validity. In the second stage, the proposed model with all hypothesised paths was tested.

5.4.1. Measurement model

5.4.1.1. Construct reliability and validity. To evaluate composite reliability (CR), Cronbach's alpha (α) and average variance extracted (AVE) were tested based on the data of unremoved scale items. As Table 5 shows, the CR values for 14 latent constructs were above the threshold value 0.70 (Anderson & Gerbing, 1988), ranging from 0.819 (infrastructure) to 0.985 (awareness). Cronbach's α values were also similar to these recorded for CR, with all constructs higher than 0.70 as recommended by Nunnally (1978). The lowest value in this regard was for infrastructure (Cronbach's α = 0.818), while the highest value was for awareness (Cronbach's α = 0.982). Furthermore, all 14 latent constructs had AVE values higher than the cut-off value of 0.50 (Fornell & Larcker, 1981; Hair Jr., Black, Babin, & Anderson, 2010). As Table 5 shows, the lowest value of AVE was for infrastructure (0.601), while the largest value was for awareness (0.956).

The standardised regression weights largely supported the scale items. Apart from five items (TRM4; TRM5; AWR4; EXP7; ENG4), all scale items loaded on their latent constructs with standardised regression weight values not less than 0.50, as shown in Table 6. Together with the results reported in Table 5, the confirmatory factor analysis provides further evidence supporting the convergent validity of the latent constructs used in the current study. This approach is highly recommended by numerous SEM scholars (e.g., Byrne, 2010; Fornell & Larcker, 1981; Hair Jr. et al., 2010; Tabachnick & Fidell, 2007).

The final validity test was for discriminant validity, the results for which are presented in Table 7. The square root of AVE values for each factor was above the estimates of correlations with all corresponding factors. This again provides further support for the scale validity used in the current study.

5.4.1.2. Model fit. The goodness of fit of the model was initially tested for all 51 scale items. Several highly suggested fit indices (goodness-of-fit index [GFI]; adjusted goodness-of-fit index [AGFI]; comparative fit index [CFI]; normed chi-square [CMIN/DF]; normed-fit index [NFI]; and root mean square error of approximation [RMSEA]) were inspected to see if the measurement model adequately fit the observed data

Table 5
Construct reliability and validity.

Construct	CR	Cronbach's α	AVE
Technology Roadmapping (TRM)	0.896	0.892	0.741
Professional Expertise (PRE)	0.891	0.880	0.733
Attitude (ATT)	0.888	0.888	0.615
Infrastructure (INFR)	0.819	0.818	0.601
Technicality (T.C)	0.868	0.865	0.625
Awareness (AWR)	0.985	0.985	0.956
SMEs' AI Practices (P.AI)	0.944	0.947	0.810
Service Experience (EX)	0.934	0.934	0.704
Customer Engagement (ENG)	0.847	0.836	0.603
Joint Planning (R.GP)	0.915	0.914	0.782
Joint Problem Solving (R.GPS)	0.905	0.923	0.763
Financial Performance (FIN)	0.927	0.925	0.809
Non-Financial Performance (N-F)	0.912	0.911	0.776

Table 6
Standardised regression weights.

Item		Construct	Estimate
TRM1	←	TRM	0.795
TRM2	←	TRM	0.914
TRM3	←	TRM	0.870
AWR1	←	AWR	0.960
AWR2	←	AWR	0.992
AWR3	←	AWR	0.981
ATT1	←	ATT	0.769
ATT2	←	ATT	0.863
ATT3	←	ATT	0.845
ATT4	←	ATT	0.750
ATT5	←	ATT	0.679
T.C1	←	T.C	0.789
T.C2	←	T.C	0.916
T.C3	←	T.C	0.777
T.C4	←	T.C	0.659
PRE1	←	PRE	0.791
PRE2	←	PRE	0.960
PRE3	←	PRE	0.807
INFR3	←	INFR	0.730
INFR2	←	INFR	0.776
INFR1	←	INFR	0.818
P.AI1	←	P.AI	0.771
P.AI2	←	P.AI	0.864
P.AI3	←	P.AI	0.959
P.AI4	←	P.AI	0.990
ENG1	←	ENG	0.870
ENG2	←	ENG	0.906
ENG3	←	ENG	0.844
EX1	←	EX	0.822
EX2	←	EX	0.856
EX3	←	EX	0.786
EX4	←	EX	0.864
EX5	←	EX	0.888
EX6	←	EX	0.814
N-F1	←	N-F	0.862
N-F2	←	N-F	0.883
N-F3	←	N-F	0.897
FIN1	←	FIN	0.901
FIN2	←	FIN	0.934
FIN3	←	FIN	0.862
R.GPS1	←	R.GPS	0.773
R.GPS2	←	R.GPS	0.857
R.GPS3	←	R.GPS	0.978
R.GP1	←	R.GP	0.843
R.GP2	←	R.GP	0.910
R.GP3	←	R.GP	0.898

(Byrne, 2010; Hair Jr. et al., 2010; Tabachnick & Fidell, 2007). However, as shown in Table 8, several fit indices (i.e., CMIN/DF = 4.014; GFI = 0.781; AGFI = 0.722; and NFI = 0.814) were not within their recommended levels. Therefore, the measurement model was revised by dropping the items with factor loadings below 0.50 as recommended by Byrne (2010), namely TRM4, TRM5, AWR4, EXP7, ENG4. Table 8 shows

Table 7
Discriminant validity.

	FIN	TRM	AWR	ATT	INFR	PRE	T.C.	EX	R.GP	P.AI	R.GPS	ENG	N-F
FIN	0.899												
TRM	0.457	0.861											
AWR	0.572	0.651	0.978										
ATT	0.459	0.246	0.409	0.784									
INFR	0.485	0.292	0.366	0.533	0.775								
PRE	0.363	0.200	0.286	0.473	0.427	0.856							
T.C	−0.066	−0.009	−0.032	−0.118	−0.113	0.029	0.791						
EX	0.733	0.595	0.606	0.520	0.450	0.365	−0.063	0.839					
R.GP	0.777	0.482	0.538	0.494	0.493	0.357	−0.094	0.752	0.884				
P.AI	0.659	0.592	0.744	0.518	0.500	0.357	−0.053	0.706	0.617	0.900			
R.GPS	0.650	0.575	0.701	0.510	0.467	0.344	−0.050	0.720	0.598	0.841	0.873		
ENG	0.681	0.628	0.721	0.411	0.363	0.290	−0.029	0.739	0.642	0.710	0.701	0.777	
N-F	0.820	0.484	0.557	0.451	0.474	0.327	−0.053	0.733	0.770	0.627	0.613	0.632	0.881

Note: **Bold** indicates square root of average variance extracted.

that the new fit indices for the revised version of the measurement model were within the acceptable levels (CMIN/DF = 2.841; GFI = 0.907; AGFI = 0.845, RMSEA = 0.062; NFI = 0.921; CFI = 0.952).

5.4.2. Structural model results

The second stage of the SEM analysis of the model largely supported the proposed conceptual model. All fit indices of the structural model were within their suggested values (CMIN/DF = 2.901; GFI = 0.903; AGFI = 0.821, RMSEA = 0.067; NFI = 0.919; CFI = 0.951). Furthermore, the model accounts for adequate variance (R^2) in the main endogenous factors (SMEs' AI practices = 0.69; service experience = 0.59; customer engagement = 0.56; joint problem solving = 0.70; joint planning = 0.45; financial performance = 0.50; non-financial performance = 0.48). The model hypotheses were tested based on the path coefficient results. As Table 9 shows, apart from two hypothesised paths (T.C → P.AI; PRE → P.AI), all paths were found to be significant. As for the main P.AI antecedents, awareness was the most influential factor contributing to SMEs' AI practices ($\gamma = 0.452, p < 0.000$), followed by technology roadmapping ($\gamma = 0.228, p < 0.000$). Attitude ($\gamma = 0.188, p < 0.000$) and infrastructure ($\gamma = 0.167, p < 0.000$) were also found to have a significant positive effect on SMEs' AI practices. However, neither technicality nor professional expertise were able to significantly predict variance in SMEs' AI practices. Path coefficient results support the significant impact of SMEs' AI practices on all measured consequences. In detail, joint problem solving ($\gamma = 0.784, p < 0.000$), service experience ($\gamma = 0.768, p < 0.000$), and customer engagement ($\gamma = 0.759, p < 0.000$) are most influenced by SMEs' AI practices. Another strong hypothesised path exists between SMEs' AI practices and joint planning ($\gamma = 0.669, p < 0.000$). Both dimensions of business performance, financial ($\gamma = 0.709, p < 0.000$) and non-financial ($\gamma = 0.681, p < 0.000$) performance, were also found to be significantly predicted by the role of SMEs' AI practices.

6. Discussion

Given the importance of SMEs and their vital role in economic

Table 8
Model goodness of fit.

Fit indices	Cut-off point	Measurement model (original)	Measurement model (revised)
CMIN/DF	≤3.000	3.451	2.841
GFI	≥ 0.90	0.781	0.907
AGFI	≥ 0.80	0.722	0.885
NFI	≥ 0.90	0.814	0.921
CFI	≥ 0.90	0.932	0.952
RMSEA	≤ 0.08	0.071	0.062

Table 9
Results of hypotheses testing.

Hypothesis	Hypothesised path	Estimate	SE	CR	p
H1a	TRM → P.AI	0.228	0.040	4.714	***
H1b	PRE → P.AI	0.043	0.036	1.170	0.242
H1c	ATT → P.AI	0.188	0.051	4.196	***
H2a	INFR → P.AI	0.167	0.060	3.721	***
H2b	T.C → P.AI	0.002	0.034	0.078	0.937
H2c	AWR → P.AI	0.452	0.038	9.616	***
H3a	P.AI → EX	0.768	0.049	14.550	***
H3b	P.AI → ENG	0.759	0.053	14.925	***
H4a	P.AI → R.GP	0.669	0.054	12.850	***
H4b	P.AI → R.GPS	0.784	0.054	14.518	***
H5a	P.AI → FIN	0.709	0.054	14.047	***
H5b	P.AI → N-F	0.681	0.054	13.155	***

Note: TRM = technology roadmapping; PRE = professional expertise; ATT = attitude; INFR = infrastructure; T.C = technicality; AWR = awareness; P.AI = SMEs' AI practices; EX = service experience; ENG = customer engagement; R.GP = joint planning; R.GPS = joint problem solving; FIN = financial performance; N-F = non-financial performance.

sustainability, especially in developing countries, this study has investigated how the implementation of AI can improve B2B SME performance in Saudi Arabia. The study has successfully provided scientific evidence for understanding the antecedents and consequences of AI acceptance by B2B SMEs. According to the structural model results, the main antecedents of AI practices have a high predictive power: the significant AI enablers and AI readiness factors account for 69% of variance in B2B SMEs' acceptance of AI practices. Furthermore, considerable variance was also found in the main consequences of acceptance of AI practices (service experience = 59%; customer engagement = 59%; joint planning = 45%; and joint problem solving = 70%; financial performance = 50%; non-financial performance = 48%).

According to the coefficient results, two dimensions of technology readiness (awareness and technical infrastructure) were found to be key determinants of B2B SMEs' acceptance of AI practices. This result indicates that AI practices have more potential to be successfully utilised if adequate knowledge and information about the features and benefits of AI applications have been previously captured by B2B SMEs (Delponte & Tamburrini, 2018). Technical infrastructure as another dimension of technology readiness was also found to be a key prerequisite for acceptance of AI practices. Thus, B2B SMEs are more likely to successfully implement AI practices if the required technical resources and facilities are available. This could be attributed to the particular nature of AI practices, which depend on the availability of highly advanced networks and systems (Gusev, 2020). This challenge would play a more crucial role in B2B SMEs due to the limited financial resources that SMEs have to buy or rent such infrastructure (Amaral & Peças, 2020). Nevertheless, technicality was not able to predict any variance in the acceptance of AI practices, which means that B2B SMEs in Saudi Arabia, when making their decision to implement AI practices, are not concerned about the ease and time required to become familiar with AI practices. This is similar to findings by Asare et al. (2012) that SMEs did not consider lack of technical know-how to be a hindrance to adoption of ICT. People are more engaged with smart technologies, so they have more familiarity with and knowledge of how to use the technology than ever before. Moreover, people are more likely to override any limitations regarding the level of complexity if they perceive that the new system is more valuable and can contribute to improved performance and productivity. This assumption was recently confirmed by Sohn and Kwon (2020) who noticed that technicality perceived in AI-based intelligent products is less important than the utilitarian and hedonic value of these products.

SMEs' AI enablers (i.e., attitude and technology roadmapping) were also confirmed to play a crucial role in facilitating the acceptance of AI practices by B2B SMEs. Such results indicate the importance of a well-informed and clear roadmap for any process of technology

transformation, which would not succeed based on a random decision-making process. Smart technologies such as AI practices require a strategy to guide decision-makers in SMEs to carry out appropriate actions that serve this aspiration (Borges, Laurindo, Spínola, Gonçalves, & Mattos, 2020). This, in turn, calls for the presence of technology roadmapping that clearly identifies the main goals, objectives, and decisions required for any implementation programme of AI practices. Practically, such a roadmap provides guidelines to help B2B SMEs make the right decisions about AI utilisation, and it also represents a solid basis for addressing the main issues that could arise prior to or during the technology implementation process (Jeffrey, Sedgwick, & Robinson, 2013). The results regarding the significant role of technology roadmapping yielded in the current study are similar to those reported by Ghobakhloo and Ching (2019). By the same token, AI practices would not be successful without managers, owners, and employees having positive thoughts, beliefs and feelings about AI practices. Attitudes have been largely identified as key determinants of people's intention and reactions towards new systems. The current study's result regarding the significant impact of attitude is very close to those reported by Cao et al. (2021).

The results of this study disprove the significant impact of professional expertise in the acceptance of AI practices by B2B SMEs in Saudi Arabia. This means that the levels of skills and technical knowledge do not present critical issues from the perspective of B2B SMEs when forming their decision to adopt AI practices. This could be due to the increasing level of awareness of AI systems, both globally and in Saudi Arabia, which is suggested by the results relating to the awareness construct in the current study. Specifically, awareness items in the questionnaire were all highly rated by participants, with an average mean of 3.72. This suggests that there is extensive awareness of AI practices in Saudi Arabia. Another explanation for the non-significant influence of professional expertise could be that AI practices are less complicated and do not require much effort or human intervention. Consequently, there is less need for training by experts.

The results for the path coefficients largely support the significant role of acceptance of AI practices in contributing to both dimensions of customer AI-based interactions (business customer engagement and service experience with AI). Due to its interactive and smart nature, AI can capture, analyse, and organise a huge amount of data and make it quickly available at the time a customer needs or requests. Moreover, as discussed by Prentice and Nguyen (2020) and Morosan and Bowen (2018), AI tools are more able to work reliably and provide an outstanding customer service experience in a consistent manner. This, in turn, will be reflected in the level of confidence and trust that business customers have in an AI-based customer service. In addition, AI practices enable business customers to have a highly customised service experience that matches their needs and expectations because such practices enjoy more flexibility that facilitates SMEs in modifying their marketing tactics and actions according to customers' preferences. The results regarding the impact of AI practices on service experience with AI are close to those of previous studies (Ameen et al., 2021; Chen et al., 2020; Haenlein & Kaplan, 2019; Morosan & Bowen, 2018; Prentice et al., 2020; Prentice & Nguyen, 2020). As for business customer engagement, a very strong causal path was noticed with acceptance of AI practices. This means that the adoption of AI practices creates more opportunities for B2B SMEs to be actively engaged with their targeted business customers. As discussed regarding AI-based customer service, AI practices can create an interactive and communicative environment capable of linking the company with its customers and promoting synchronous contact between them, regardless of any temporal or spatial obstacles. Furthermore, AI practices can positively contribute to the customer service experience with SMEs, which, in turn, creates an appropriate environment that enhances opportunities for integration and engagement among business partners. Several studies have demonstrated the impact of AI practices on customer engagement (e.g., Prentice et al., 2020; Prentice & Nguyen, 2020).

The impact of acceptance of AI practices on both dimensions of

relational governance (joint planning and joint problem solving) was supported, as proposed by the conceptual model. This means that the vast majority of participants positively perceived AI practices as levers helping B2B SMEs to jointly plan their future activities (e.g., future volume demands and sharing of long-term strategy) as well as to solve all problems that may occur. Compared with traditional communication tools, AI practices create more opportunities for business partners to communicate and interact with each other effectively; thus, business partners are more able to mutually collaborate, share knowledge, and co-create value (McAfee et al., 2012; Mogaji et al., 2020; Prentice & Nguyen, 2020; Rahman et al., 2021). In this regard, Dwivedi, Hughes, et al. (2021) also argued that business partners are more able to effectively validate large amounts of data when preparing their long-term plans, especially those relating to supply-and-demand issues.

Both financial performance and non-financial performance were found to be considerably predicted by the role of AI practices. These results clearly indicate the extent to which B2B SMEs in Saudi Arabia are convinced of the role and value of AI practices for their business performance. Practically, B2B SMEs' ability to make good decisions, and hence to have desirable performance outcomes (e.g., competitive advantage, customer retention, market share, sales volume, and profitability), largely depends on their understanding of their customers' behaviours, attitudes, and experience (Duan, Edwards, & Dwivedi, 2019). According to Demirkan and Delen (2013), AI practices have been increasingly receiving attention due to their ability to create and sustain a company's competitive advantage. This could also be because AI practices significantly contribute to the ability of business organisations to innovate and develop products. Thus, customers are more satisfied about what is introduced by these organisations, which accelerates growth and volume of sales. Furthermore, AI practices have empowered the innovation process of new products and services by making it both more cost-effective and less risky (Dubey et al., 2020). This is due to the ability of AI-empowered practices to collect and process large amounts of diversified data, thereby facilitating better decisions relating to customer demand, supply chain management, and organisation capacity (Basri, 2020; Dubey et al., 2019; Dubey et al., 2020; Rahman et al., 2021). Furthermore, the results regarding engagement and service experience largely support AI as a crucial mechanism in helping SMEs and their business partners to adapt their long-term relationship and resources in a way that sustains their business performance (Nishant, Kennedy, & Corbett, 2020).

6.1. Theoretical implications

This study has been conducted due to the absence of a clear and comprehensive picture regarding the antecedents and consequences of AI practices from the perspective of B2B SMEs. It has theoretically proposed an inclusive model and empirically tested the main hypothesised paths among the antecedents and consequences of AI practice acceptance based on data obtained from 392 respondents from B2B SME organisations in Saudi Arabia. Accordingly, this study provides a broader base for theoretical and practical understanding of issues relating to AI practices in SMEs and the B2B sector in general.

This study operationalised new mechanisms from the TOE theory by considering SMEs' AI enablers; accordingly, it has extended the theoretical horizon of TOE. For example, it presents more evidence of technology roadmapping as one of the main AI enablers of a successful implementation of AI practices by SMEs. Another theoretical contribution is the validation of the role of SMEs' AI readiness. In this regard, this research has shed light on variables (AI infrastructure and AI awareness) that have long been in need of further study and investigation.

With regard to the consequences of AI practices, this study has successfully gone further than the acceptance stage by addressing customer AI-based interaction in terms of service experience with AI and customer engagement. This clearly shows that the success of AI practices concerns not only their implementation but also their ability to enhance

opportunities to improve customer service and engagement. A significant contribution of this study is its validation of a strong relationship between AI practices and factors of relational governance, which has rarely been discussed in the area of AI for SMEs. Thus, this study has broadened the scope of AI acceptance in terms of business customer engagement and the effectiveness of the service delivery process, both of which contribute to business performance and relational governance (Basri, 2020). As discussed in the literature review section, there is a dearth of literature on the AI practices of B2B SMEs, particularly SMEs in Arab Middle Eastern countries such as Saudi Arabia. Accordingly, this research enhances the scholarly understanding of AI practices in Saudi B2B SMEs.

6.2. Managerial implications

As discussed above, this study was conducted with the aim of helping B2B SMEs in Saudi Arabia to better understand how they can successfully implement AI practices. Therefore, the empirical results constitute a frame of reference for the most important dimensions that must be considered to ensure the successful implementation of AI practices. One of the most important dimensions is the necessity of B2B SMEs having technology roadmapping. This initially requires B2B SMEs to consciously and accurately prepare a clear roadmap for the implementation and use of contemporary technologies. It is strongly recommended that business partners be involved in all stages of designing and developing technology roadmapping. Such a technology roadmap should carefully identify the short- and long-term objectives of a digital transformation process (Jeffrey et al., 2013). More importantly, the technology roadmapping should be moved from its theoretical framework to its practical framework so that it can serve as a guide for all decisions and actions pertaining to AI practices.

The results also provide clues about the importance of attitudes for the successful acceptance of AI practices. This requires SMEs to further explore the nature of such attitudes to AI practices and then to shape such attitudes. Therefore, a survey could be conducted to discover the common opinions and attitudes of stakeholders towards AI practices. In parallel, more internal marketing efforts should be made to improve and sustain positive feelings and thoughts that AI practices can provide SMEs with more control, convenience, and safety in their operations.

In relation to AI readiness, awareness and infrastructure should receive attention from B2B SMEs in their endeavours to implement AI practices. It is important to extensively familiarise people with AI practices and increase awareness of their benefits and characteristics. More education and training programmes can help to raise the current level of awareness and the value generated from AI practices. According to Lin and Lin (2008), B2B SMEs should ensure that AI vendors provide more assistance by educating and training their employees in the skills and technical knowledge required for effective use of AI practices. Social media platforms (e.g., LinkedIn, Twitter, YouTube, and TEDx) could also be useful when creating public awareness among stakeholders in the B2B SME sector (Dwivedi, Ismagilova, Hughes, Carlson, et al., 2021).

With regard to infrastructure, it is important that B2B SMEs either have or invest in the facilities and technical resources needed to successfully implement AI practices. Therefore, B2B SMEs that hope to utilise AI practices in their business model should initially evaluate their existing technical capabilities and resources, thereby enabling them to identify which areas of technical infrastructure need investment and improvement (Lin & Lin, 2008). Further, B2B SMEs should concentrate on their ability to integrate AI practices with current digital applications and infrastructure.

6.3. Limitations and future research directions

Although this study has provided several theoretical and practical implications, some limitations could be addressed in future research. First, this study is cross-sectional, where both antecedents and

consequences of AI practices were tested at one point in time. However, this does not provide a full picture of the long-term applicability of the current study's results. Therefore, a longitudinal study is recommended to investigate the impact of the current study's factors prior to, during, and after implementation of AI practices. Second, this study examined the related issues of AI practices only from the supplier's perspective within the B2B SME sector, thus restricting the ability to expand the generalisability of the results to the business customer's perspective. Therefore, future studies could consider this phenomenon from both perspectives: SMEs as both vendors and business customers. Third, despite the current study model being based on the TOE model, other macro-environmental factors (e.g., culture, government regulations, national technological advancement level, and economic situation) were not considered in the current study. Thus, it would be useful for future researchers to investigate whether macro-environmental factors positively or negatively impact the successful implementation of AI practices. Finally, the dark side of AI (Cao et al., 2021), for example issues of privacy, security, and ethics were not addressed in the current study, so future studies should consider how these affect the tendency of B2B SMEs towards AI practices.

7. Conclusion

The goal of this paper has been to evaluate the main antecedents and consequences of acceptance of AI practices by B2B SMEs in Saudi Arabia. By reviewing previous work on AI within the SME sector, it was found that there is a lack of understanding regarding this topic and that only a few studies have addressed it. The TOE model was considered as a theoretical basis for the model proposed by this study. Specifically, SMEs' AI-enablers (i.e., AI-based technology roadmapping, professional expertise, and attitude) and SMEs' AI readiness (i.e., infrastructure, technicality, and awareness) were proposed as key antecedents of acceptance of AI practices. The conceptual model also proposed the consequences of AI practices in three main categories: relational governance, customers' AI-based interactions, and SMEs' AI-enabled performance. Empirical data were collected from participants who own or work in B2B SMEs. The SEM results support the current study model fitness and predictive validity, as well as the majority of the proposed research hypotheses, with the exceptions being the impact of both technicality and professional expertise on acceptance of AI practices. Solid theoretical and logical justifications for the results have been discussed. In addition, the main theoretical and practical implications of this study have been outlined.

Appendix

Construct	Items	Reference
Technology Roadmapping	TRM1 Our company has a clear and effective process of developing AI-based technology roadmapping.	Jeffrey et al. (2013); Lee, Kim, and Phaal (2012)
	TRM2 Our AI-based roadmap has clear, defined goals, and prioritised objectives to avoid trying to do too much.	
	TRM3 AI-based TRM output is sufficiently utilised in order to make a decision on a new technology R&D project.	
	TRM4 AI-based TRM output is sufficiently utilised during the process of a new technology R&D project.	
	TRM5 AI-based TRM output is actively referred to by our organisation/team for addressing major issues that occur in the process of a new technology R&D project.	
Infrastructure	INFR1 Our firm has a good AI-based infrastructure including all IT components.	Lin and Lin (2008)
	INFR2 There are integrated Web applications encompassing different functional areas.	
	INFR3 Our firm shares the databases for various applications, rather than having a separate database for each application.	
Professional Expertise	PRE1 IT/IS employees are generally aware of the functions of e-business and AI applications.	
	PRE2 Our firm hires highly specialised or knowledgeable personnel for e-business and AI practices.	
	PRE3 IT/IS employees are well trained in e-business and AI applications.	
Attitude	ATT1 AI gives our company more control of daily operations.	Kros, Glenn Richey, Chen, and Nadler (2011)
	ATT2 Processes and equipment that use the newest AI technologies/applications are much more convenient to use.	
	ATT3 Other people come to us for advice on new technologies/applications e.g., AI.	
	ATT4 In general, we are among the first in our industry to acquire new technology e.g., AI.	
	ATT5 We can usually implement AI-based high-tech products without the help of others.	
Awareness	AWR1 We know about AI applications.	Alsheikh and Bojei (2014)
	AWR2 We have received enough information about the benefits of using AI applications.	
	AWR3 We have received enough information on how to use AI applications.	
	AWR4 We have received information about the security system of AI applications.	
Technicality	T.C1 It would be easy to use the AI applications technically.	Sohn and Kwon (2020)
	T.C2 It would be easy to operate the AI applications.	
	T.C3 It would not take much time to become familiar with the AI applications.	
	T.C4 It does not look difficult to use the AI applications.	
SMEs' AI Practices	P.AI1 AI is a mainstream technology that will be dominant in our industry.	Kros et al. (2011)
	P.AI2 AI is well accepted by the top management in our company.	
	P.AI3 AI is well accepted by most of the frontline employees in our company.	
	P.AI4 AI is or will be the core technology critical to our company's future success.	
Service experience	EX1 AI tools provide our business customers with information in a timely fashion.	Prentice and Nguyen (2020)
	EX2 AI tools return answers to requests from our business customers quickly.	
	EX3 AI tools operate reliably.	
	EX4 The operation of AI tools is dependable.	
	EX5 AI tools can flexibly adjust to new demands or conditions.	
	EX6 AI tools are versatile in addressing needs.	
	EX7 There are few errors in the information I obtain from AI tools.	
Customer Engagement	ENG1 Using AI will enhance information communication with business customers.	Dessart, Veloutsou, & Morgan-Thomas, 2015; Mollen and Wilson (2010).
	ENG2 AI technology enables professionals to identify and connect to current and prospective business customers.	
	ENG3 AI is useful for dealing with complaints and after-sales follow up.	

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(continued)

Construct	Items	Reference
Joint Planning	ENG4 AI applications enhance the effectiveness of CRM process and capabilities.	Claro et al. (2003)
	R.GP1 AI applications will help us and our business customers to plan volume demand for the next season together.	
	R.GP2 AI applications will help us and our business customers to plan the various demands for the next season together with our major supplier.	
Joint Problem Solving	R.GP3 AI applications will enable us and our business customers to share our long-term plans for our products.	Diffley and McCole (2015); Hernaus et al. (2012); Jain et al. (2014)
	R. AI applications will enable us and our business customers to jointly deal with problems that arise in the relationship.	
	GPS1 AI applications will enable us and our business customers to jointly take responsibility for getting things done.	
	R. AI applications will strengthen our relationship and commitment to our business customers.	
Non-Financial Performance	GPS2	Diffley and McCole (2015); Hernaus et al. (2012); Jain et al. (2014)
	GPS3	
	N-F1 AI applications will help us to gain strategic advantages over our competitors.	
Financial Performance	N-F2 AI applications will help us to retain existing clients and to attract new ones.	Jain et al. (2014)
	N-F4 Using AI applications will increase the levels of business customer satisfaction.	
	FIN1 Using AI applications will increase our market share.	
Performance	FIN2 Using AI applications will generate a high volume of sales.	Jain et al. (2014)
	FIN3 Our relationship with this supplier has helped achieve rapid growth.	

Note: *italics* = item removed following Confirmatory Factor Analysis.

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